

Smart Solar Water Heating System with Temperature control

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Abstract—The increasing cost of energy for domestic and industrial water heating has become a significant economic and sustainability challenge, particularly in regions with high electricity tariffs and growing energy demand. Conventional electric and fuel-based water heating systems are energy-intensive and contribute substantially to operational costs and carbon emissions. This paper presents the design and implementation of a smart solar water heating system with temperature control aimed at reducing energy consumption while ensuring user comfort and system efficiency. The proposed system integrates solar thermal collectors with embedded temperature sensors, a microcontroller-based control unit, and an auxiliary heating mechanism. Real-time temperature monitoring and intelligent control algorithms are employed to regulate water temperature, optimize solar energy utilization, and minimize reliance on grid electricity. Simulation results demonstrate that the system maintains the desired water temperature within predefined limits while achieving significant reductions in energy consumption compared to conventional water heating methods. The system offers a cost-effective, energy-efficient, and environmentally sustainable alternative for water heating applications, making it suitable for residential and small-scale commercial use. As for future developments, enhancing the system with predictive solar forecasting algorithms, integrating IoT connectivity for remote monitoring and control, incorporating thermal energy storage tanks for extended hot water availability during nighttime, and exploring integration with photovoltaic panels for a complete renewable energy solution could be researched.

Index Terms—solar water heating, energy efficiency, smart control, temperature regulation, renewable energy

I. INTRODUCTION

Heating water consumes a significant amount of energy in both homes and businesses, resulting in high bills and environmental impacts. Solar water heating systems offer a sustainable alternative by harnessing solar energy, but conventional setups often produce water that is excessively hot or require extra heating to reach a comfortable temperature.

This project seeks to develop an intelligent solar water heating system with built-in temperature regulation. It will efficiently collect and store solar-heated water during the day and ensure that it is delivered at the precise temperature selected by the user. As a result, there is no energy wasted from overheating and no need for additional heating sources.

The objective is to reduce the cost of water heating while improving user convenience. The system will consist of two

core components: a solar water heating unit and a temperature control module. Sensors will track the water temperature, and the system will automatically adjust the ratio of hot and cold water to provide water at the user's preferred temperature.

II. LITERATURE REVIEW

Water heating accounts for a significant portion of residential and commercial energy consumption, leading to increased operational costs and higher dependence on conventional energy sources. Several studies have highlighted that electric and fossil-fuel-based water heaters contribute substantially to peak load demand and carbon emissions, especially in developing countries where electricity tariffs are continuously rising [1].

Solar water heating systems (SWHS) have been widely researched as a sustainable alternative to conventional water heating methods. Early studies focused on flat-plate and evacuated tube collectors, demonstrating their ability to reduce electricity consumption by 50–70% under favorable climatic conditions [2]. However, traditional solar water heaters suffer from limitations such as temperature fluctuations, heat losses, and lack of adaptability to varying user demand and weather conditions [3].

To overcome these challenges, researchers have proposed integrating control mechanisms into solar water heating systems. Temperature-controlled solar water heaters utilize sensors and controllers to maintain the outlet water temperature within a desired range, improving user comfort and system efficiency. Studies have shown that incorporating thermostatic control and feedback mechanisms can significantly reduce overheating and thermal losses while enhancing system reliability [4].

Recent advancements have introduced smart solar water heating systems that combine microcontrollers, temperature sensors, and intelligent control algorithms. These systems dynamically regulate water flow, auxiliary heating, and storage based on real-time temperature data and user requirements. Research indicates that smart control strategies can further reduce energy costs by optimizing solar energy utilization and minimizing reliance on backup electric heaters [5].

Additionally, the integration of Internet of Things (IoT) technologies has enabled remote monitoring and adaptive control of solar water heating systems. IoT-based solutions allow

users to monitor temperature, energy savings, and system performance, leading to improved energy management and predictive maintenance [6]. Despite these advancements, the high initial cost and complexity of smart systems remain barriers to widespread adoption.

Based on the reviewed literature, it is evident that a smart solar water heating system with temperature control offers a promising solution to rising energy costs. However, there is a need for cost-effective, efficient, and easily implementable designs suitable for residential applications, particularly in regions with high solar potential.

III. PROBLEM STATEMENT

Conventional electric water heating systems contribute significantly in both residential and commercial energy consumption, especially in tropical countries such as Sri Lanka, where hot water demand remains high throughout the year. These systems depend heavily on grid electricity, leading to increased operational costs, higher peak demand, and greater greenhouse gas emissions. Although solar water heating systems offer a sustainable alternative by utilizing abundant solar energy, their practical performance is often limited by inefficient heat utilization, lack of intelligent control, and poor temperature regulation under varying solar irradiance and load conditions.

Existing solar water heaters commonly operate without adaptive control mechanisms, resulting in issues such as overheating during high solar insolation, insufficient water temperature during low irradiance periods, and unnecessary reliance on auxiliary electric heaters. In addition, the absence of dynamic system modeling and real-time control strategies prevents optimal integration of solar energy with backup heating, reducing overall system efficiency and reliability.

Therefore, there is a clear requirement for a smart solar water heating system that incorporates accurate thermal modeling and intelligent temperature control to ensure reliable hot water supply while minimizing auxiliary energy usage. Designing and validating such a system through simulation is essential to evaluate its performance under realistic summer climatic conditions and variable demand pattern, and to demonstrate its potential for reducing energy costs and improving energy efficiency.

IV. SYSTEM OVERVIEW

This flowchart shows a smart solar water heating system that automatically mixes solar-heated water with cold water to maintain a stable user-selected temperature. The system begins by calculating solar energy availability throughout the day, converting it to heated water temperature. A controller continuously compares this solar temperature with user desired temperature setpoint. If the solar water is hot enough, it calculates the perfect mixing ratio with 20°C cold water to reach exactly your setpoint without using electricity. If solar water is not hot enough, the system delivers all solar water and switches on an electric heater that calculates precisely how much power is needed to heat from the current temperature to your setpoint. The system monitors total energy consumption

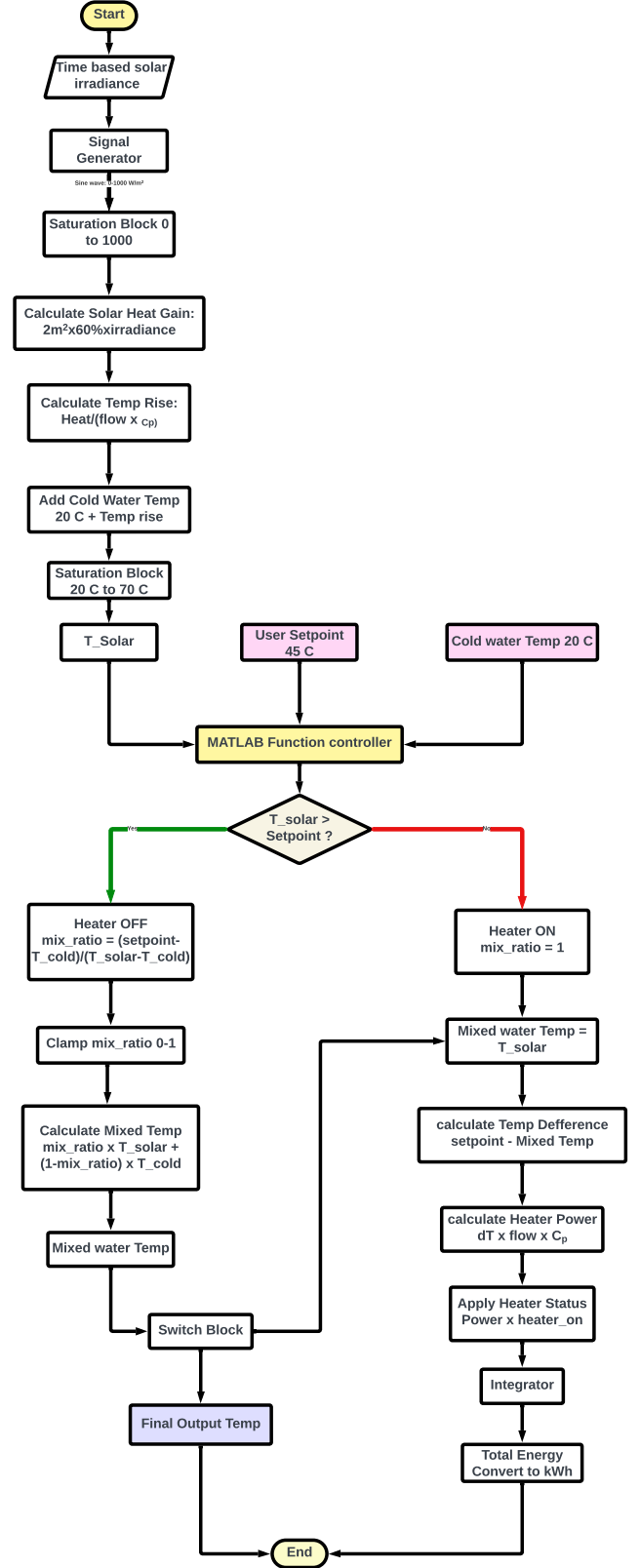


Fig. 1. Flow chart for the process

and always delivers water at user preferred temperature while maximizing solar energy use and minimizing electricity costs.

V. SYSTEM DESIGN

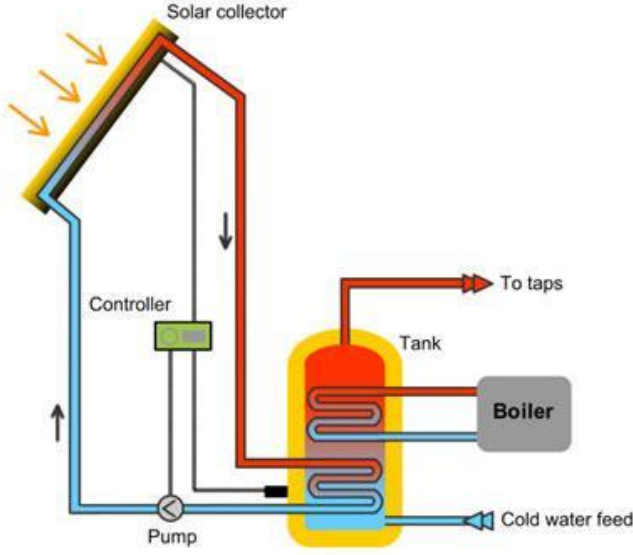


Fig. 2. System Design

A. Solar Collector

The solar collector is the primary energy-harvesting component of the system. It absorbs incident solar irradiance and converts it into thermal energy, which is transferred to the working fluid circulating through the collector tubes. The absorber surface is designed with high solar absorptivity and low thermal emissivity to maximize heat gain while minimizing losses to the environment. As the working fluid temperature increases, it is transported to the storage tank through the circulation loop. The performance of the collector depends on factors such as solar intensity, collector orientation, ambient temperature, and flow rate of the heat transfer fluid.

B. Controller

The controller functions as the central decision-making unit of the solar water heating system. It continuously monitors temperature signals from sensors located at the solar collector outlet and within the storage tank. Based on the measured temperature difference and predefined control logic, the controller regulates the operation of the circulation pump to ensure efficient heat transfer. When sufficient solar energy is available, the controller activates the pump to circulate heated fluid; otherwise, it prevents unnecessary circulation to avoid thermal losses. This control strategy improves system efficiency, protects components, and ensures stable hot water delivery.

C. Boiler

The boiler serves as an auxiliary or backup heating source to guarantee hot water availability under conditions of insufficient solar energy, such as low irradiance or high demand. It is typically integrated with the storage tank via a heat exchanger or direct heating element. When the tank temperature falls below the required setpoint, the boiler is activated to supply the additional thermal energy needed. This hybrid operation ensures uninterrupted hot water supply while minimizing electrical or fuel consumption by prioritizing solar heating whenever possible.

VI. MATLAB MODELS AND SIMULATION RESULTS

A. Solar Irradiance Stage

The solar irradiance input is created using signal generator blocks combined with a saturation block. The signal generator produces a time-varying waveform that mimics the natural pattern of sunlight throughout the day. The saturation block limits the signal to realistic values, ensuring the irradiance stays between 0 W/m^2 (nighttime) and approximately 1000 W/m^2 (maximum sunlight). This input signal is then fed into the PV (photovoltaic) array block to simulate how solar panels respond to changing light conditions.

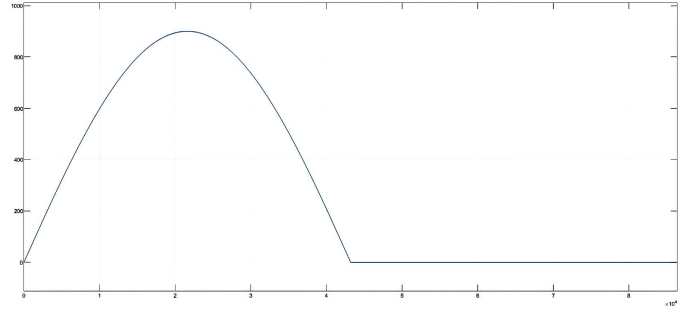


Fig. 3. Solar Irradiance Profile Over Time

The graph displays a smooth bell-shaped curve representing solar irradiance over approximately 24 hours. Starting from zero at dawn, the irradiance gradually increases during the morning hours, reaching a peak of about 900 W/m^2 around solar noon (at 12 hours). After reaching this maximum, the irradiance symmetrically decreases through the afternoon, returning to zero around 18.5 hours, simulating sunset. This pattern represents typical clear-sky conditions and serves as the driving input for the entire solar power system simulation.

B. Solar Heated Water Temperature Stage

This stage calculates the temperature of water heated by the solar panels. The solar irradiance input from the previous stage is processed through gain blocks and math function blocks that apply thermodynamic calculations. These blocks convert the solar energy (W/m^2) into thermal energy that heats the water. The gain block likely represents factors such as the solar collector's efficiency, heat transfer coefficient, and the

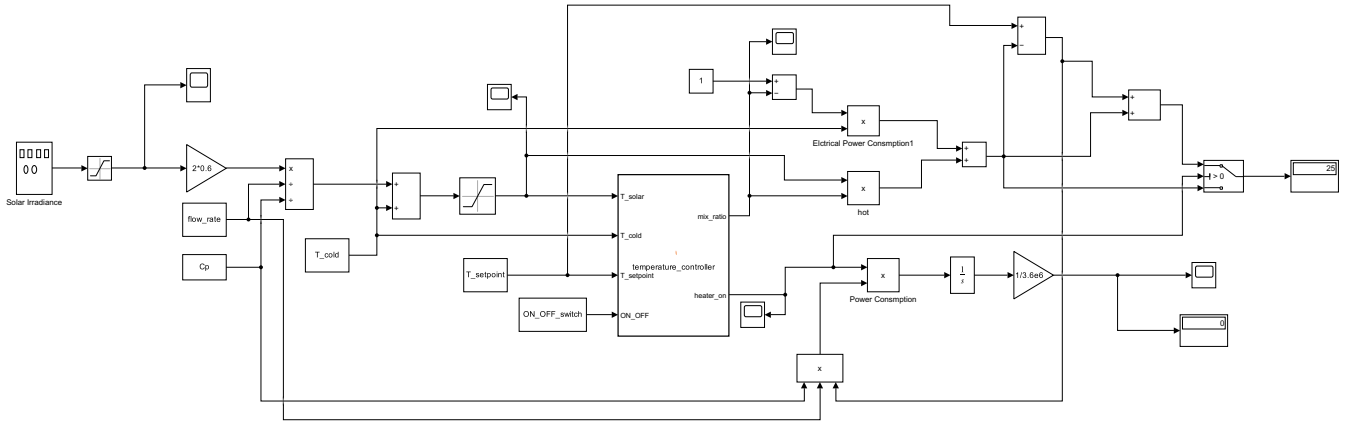


Fig. 4. Overall system architecture

specific heat capacity of water. The math function performs the necessary calculations to determine how much the water temperature increases based on the absorbed solar energy over time.

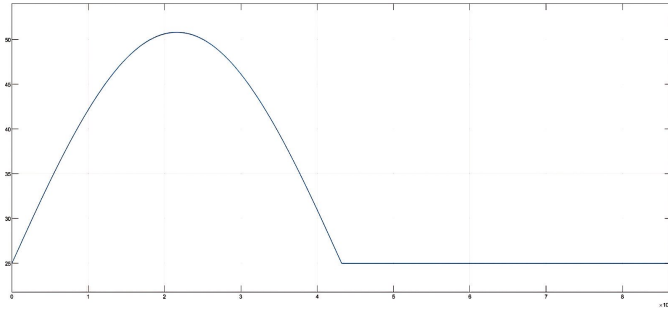


Fig. 5. Solar-Heated Water Temperature Variation

The graph shows water temperature variation following the solar irradiance pattern. Starting at approximately 25°C (ambient temperature) in the early morning, the water temperature rises as solar irradiance increases. The temperature peaks at around 50°C during solar noon (at 2 hours) when irradiance is maximum. As the sun sets and irradiance decreases, the water temperature gradually drops back to approximately 25°C by evening (around 4.5 hours). The curve demonstrates how effectively the solar thermal system converts sunlight into usable heat energy for water heating applications.

C. Heater Control Stage (On/Off Status)

This stage controls whether the electric heater needs to be activated based on the water temperature requirements. A constant block sets the user's desired temperature setpoint (T-setpoint). The temperature controller logic, implemented using a MATLAB function block, receives three inputs: T-solar (heated water temperature from solar panels), T-cold (cold water inlet temperature), and T-setpoint (desired temperature). The function calculates a mixing ratio, which determines the

proportion of hot and cold water needed to achieve the setpoint temperature. The mixing ratio is calculated as: $(T\text{-setpoint} - T\text{-cold}) / (T\text{-solar} - T\text{-cold})$. Additionally, the controller decides whether the electric heater should turn on or off based on whether the solar-heated water alone can meet the temperature demand. An on/off switch is included in the simulation to represent user control—the heater only operates when users actually need hot water, preventing unnecessary energy consumption when the system is not in use.

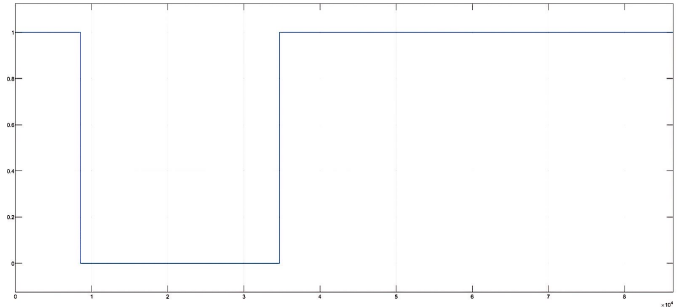


Fig. 6. Electric Heater Status (ON/OFF Control Signal)

The graph shows a binary signal representing the heater's operational status over time. The heater remains ON (value = 1) from the start until approximately 8 hour, indicating that during early morning when solar irradiance is low, the solar panels cannot heat water sufficiently to meet the setpoint temperature. At around 8 hour, the signal drops to 0 (OFF), showing that solar heating alone now provides adequate temperature. The heater stays OFF until around 16 hours when it turns back ON (value = 1) as the sun begins to set and solar heating becomes insufficient again. This control strategy ensures the heater only operates when necessary, optimizing energy efficiency by maximizing solar energy use and minimizing electric heater operation.

D. Mixing Valve Stage

This stage simulates the mixing valve that blends hot solar-heated water with cold water to achieve the desired setpoint temperature. The mixing ratio, calculated from the temperature controller in Stage 3, determines the proportion of hot and cold water needed. The first product block multiplies the mixing ratio by T_{solar} (solar-heated water temperature), calculating the contribution of hot water to the final mixture. Simultaneously, a subtract block calculates $(1 - \text{mixing ratio})$ to determine the cold water proportion. The second product block multiplies this value by T_{cold} (cold inlet water temperature), calculating the cold water contribution. Finally, an add block sums both product block outputs to produce the mixed water temperature: $T_{\text{mixed}} = (\text{mixing ratio} \times T_{\text{solar}}) + ((1 - \text{mixing ratio}) \times T_{\text{cold}})$. This mathematical operation simulates the physical mixing valve operation. This mixing process occurs when the heater is OFF, meaning the solar-heated water temperature (T_{solar}) is higher than the setpoint temperature. In this scenario, cold water must be blended in to reduce the temperature down to the comfortable setpoint level, preventing excessively hot water from being delivered to the user.

E. Electric Heater Power and Energy Consumption Stage

This stage calculates the electric heater's power consumption and total energy usage when the heater is active. A subtract block first determines the temperature difference between the setpoint temperature and the solar-heated water temperature ($T_{\text{setpoint}} - T_{\text{solar}}$), which represents how much additional heating is required. This temperature difference is then converted to required heating power (in watts or kilowatts) using gain blocks that account for factors like water flow rate, specific heat capacity, and heater efficiency. A product block multiplies this power requirement by the heater status ($\text{ON}=1, \text{OFF}=0$) from the previous stage, ensuring power consumption is only calculated when the heater is actually operating. An integrator block accumulates this power over time to calculate total energy consumed (in kWh). Finally, a gain block converts this energy into billing units for cost calculation. Both a scope and display blocks show these values for monitoring and analysis.

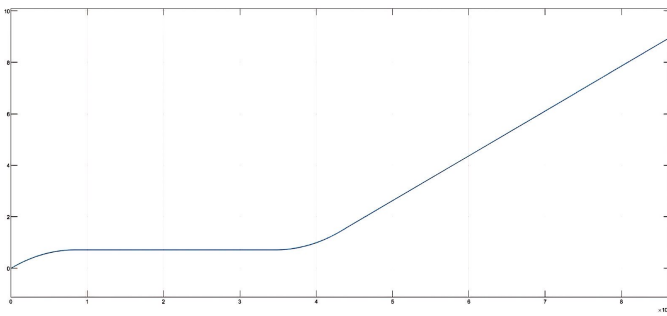


Fig. 7. Cumulative Energy Consumption by Electric Heater

The graph shows cumulative energy consumption over the simulation period. From 0 to approximately 24 hours, the curve

shows a slight gradual increase as the heater operates during early morning when solar heating is insufficient. Between 3 and 10 hours, the curve remains nearly flat because the heater is OFF—solar energy alone meets the temperature demand during peak sunlight hours. After 10 hours, the curve begins rising steeply and continuously as the heater turns back ON due to decreasing solar irradiance in the evening. The steep linear increase after 10 hours shows significant energy consumption as the system relies entirely on electric heating. By the end of the simulation period, total energy consumption reaches approximately 9 units (kWh), demonstrating the cumulative cost of supplementing solar heating with electric power during low-sunlight periods.

VII. CONCLUSION

A smart solar water heating system with temperature control has been successfully designed and simulated using MATLAB and Simulink blocks. The system demonstrates commendable energy efficiency, automatic temperature regulation, and cost-effectiveness by maximizing solar energy utilization while minimizing electric heater usage. Anticipated benefits include reduced electricity consumption, lower utility costs, decreased carbon footprint, and reliable hot water supply, leading to improved sustainability and energy independence for residential applications.

As for future developments, enhancing the system with predictive solar forecasting algorithms, integrating IoT connectivity for remote monitoring and control, incorporating thermal energy storage tanks for extended hot water availability during nighttime, and exploring integration with photovoltaic panels for a complete renewable energy solution could be researched.

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